

Chapter 72

Animating in Fusion's Spline Editor

This chapter covers how you can keyframe effects and control animations in Fusion's Spline Editor.

Contents

Spline Editor Overview	1477
Spline Editor Interface	1478
The Graph, Header, and Toolbar	1478
Renaming Splines	1480
Changing Spline Colors	1480
Navigating Around the Spline Editor	1480
Markers	1481
Autosnap	1484
Creating Animation Splines	1484
Animating with Different Spline Types	1485
Working with Keyframes and Splines	1487
Adding Keyframes	1487
Locked and Unlocked Controls Points	1488
Selecting, Moving, and Deleting Keyframes	1489
Showing Key Markers	1490
Copying and Pasting Keyframes	1490
Time and Value Editors	1492
Modifying Spline Handles	1493
Reducing Points	1494
Filtering the Spline Editor	1495
Working with Filters	1496
Changing a Spline's Status	1497
Selection States	1498

Reshaping Splines Using the Toolbar	1499
Interpolation	1499
Reversing Splines	1501
Looping Splines	1501
Time Stretching	1504
Shape Box	1505
Ease In/Out	1506
Importing and Exporting Splines	1507

Spline Editor Overview

The Spline Editor is the main area where animation is manipulated and refined. You primarily use the Spline Editor to show the changing values of parameters over time in the form of splines. Whereas Keyframes explicitly set the value of a parameter on a given frame, Splines are lines or curves that interpolate the values between keyframes. Once you set a keyframe on a parameter, a spline is created and can be displayed in the Spline Editor so you can make further refinements to the animation. However, the Spline Editor is more advanced than a standard curve editor since it can also display functions, which may not be splines, like changing characters within a text string or mathematical expressions that drive your animations.



The Spline Editor with three animated parameters

What Are Splines?

All animation uses splines that describe the value of a parameter at any given point in time. The Spline Editor graph shows the time of the comp along the horizontal axis and the value of the parameter along the vertical axis.

The advantage of using splines to represent animation instead of keyframes as in the Keyframes Editor is that splines allow you to manipulate the interpolation between keyframes. For example, if a keyframe is set at a value of 1.0 on a parameter for frame 1, followed by a keyframe value of 10.0 for frame 10, the values between keyframes are smoothly interpolated or calculated based on the shape of the spline. Using the functions and controls in the Spline Editor, you have a fantastic amount of control over that interpolation.

Spline Editor Interface

The Spline Editor is not visible by default, but you can show it at any time by clicking the Spline button in the user interface toolbar. You can also display the Spline Editor by right-clicking on a node in the Node Editor or a segment in the Keyframes Editor and choosing Edit Splines from the drop-down menu.

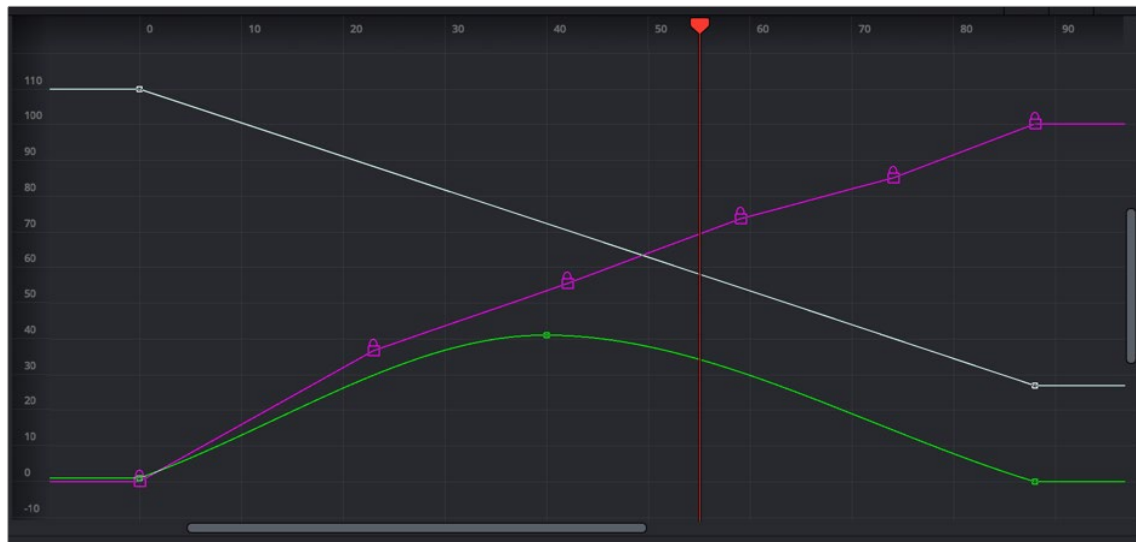
The Spline Editor can be open alongside the Node Editor or Keyframes Editor, or displayed separately in order to take up the entire work area.

The Graph, Header, and Toolbar

The Spline Editor has three main working areas: the graph, header, and toolbar. On the left side of the Spline Editor is the header, which shows a list of animated parameters. The majority of the panel is taken up by the splines displayed in the graph area, and a toolbar runs along the bottom, providing a variety of ways to manipulate the splines.

Graph

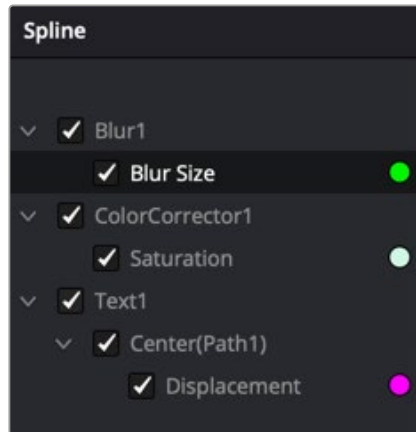
The graph is the largest area of the interface. It is here that you see and edit the animation splines. There are two axes in the graph. The horizontal axis represents time, and the vertical axis represents the spline's value. A thin bar, called the playhead, runs vertically through the graph to represent the current time as it does in the Timeline Editor. You can drag the playhead to change the current time, updating the frame displayed in the viewers.



Removing a spline from a parameter removes the animation

Spline Editor Header

The header provides a mechanism for determining what splines are visible in the graph. It shows the name of each spline in the project beneath the tool that contains that parameter. The checkbox beside each name shows whether that spline is currently displayed in the graph and whether the spline can be edited.



The Spline header

Spline Editor Toolbar

The toolbar across the bottom of the Spline Editor represents the most common operations applied to an animation spline. The various operations represented in the toolbar are all accessible from the graph's context menu as well, but the following buttons provide a faster shortcut.



Spline toolbar buttons

Playhead

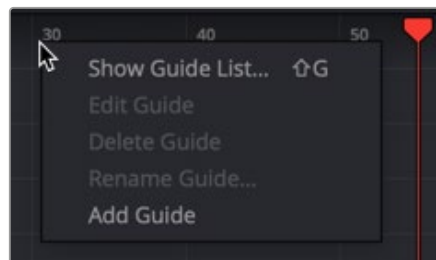
The playhead is the thin red vertical bar that runs vertically through the Spline Editor graph and represents the current time of the comp. You can drag the playhead to change the current time.

Status Bar

The status bar in the lower-right corner of the Fusion window regularly displays information about the position of the pointer, along with the time and value axes.

Contextual Menus

There are two contextual menus accessible from the Spline Editor. The Spline contextual menu is displayed by right-clicking over the graph, while the Guide contextual menu is displayed by right-clicking on the Time Ruler above the graph.



Right-click on the horizontal axis Time Ruler for the Guide menu

Renaming Splines

The name of a spline in the header is based on the parameter it animates. You can change the name of a spline by right-clicking on it in the header and choosing Rename Spline from the contextual menu.

Changing Spline Colors

Each spline in the graph is assigned a different color, making individual splines easier to identify when multiple splines are visible at once. When the spline is active, a round color swatch is displayed next to the spline's name in the header.

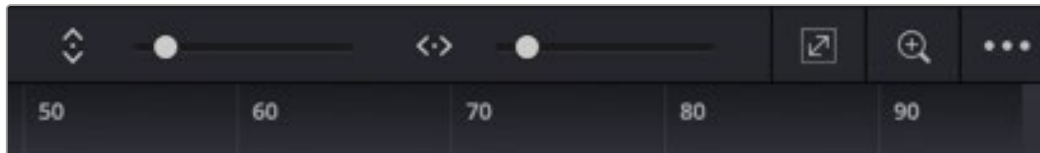
To change the color of a spline:

- 1 Click on the color circle next to the spline name in the header.
- 2 Or right-click on the name of the spline in the header and choose Change Color from the contextual menu.
- 3 Select the new color from the dialog box that appears and click OK.

Navigating Around the Spline Editor

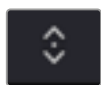
It is often necessary to magnify and pan around the graph area to ensure that the splines you want to work on are visible. In general, scaling and panning the Spline Editor works the same as in all navigable parts of the Fusion interface. However, there are several unique functions to the Spline Editor for controlling your view based on the height, width, and selection of multiple animation splines.

The most obvious navigation methods to use are the scale sliders and buttons located in the upper left of the Spline Editor panel.

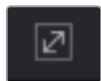


The Zoom Height and Zoom Width sliders, Fit button, and Zoom to Rectangle button can be used to navigate around the graph

To scale and pan using the sliders and buttons:



Zoom Height and Zoom Width sliders let you change the height and width of the graph area.



The Fit button attempts to rescale the view so that all currently active splines fit within the graph.



The Zoom to Rectangle button (Command-R) allows you to draw a bounding box around the area of the graph you want centered and scaled.

To scale using the axis labels:

- Place the mouse pointer over the rulers for the horizontal or vertical axis and drag to resize the graph on that axis only. The view is scaled centered on the original position of the pointer on the ruler.



Dragging in the Time Ruler scales the graph horizontally

To scale and pan with the mouse and/or keyboard:

- With the Spline Editor active, press the + and - keys on your keyboard to zoom in and out of the graph.
- You can also zoom to a specific control point by holding down the Command key and scrolling the middle mouse wheel. The mouse pointer location determines the area that gets magnified.
- Position the mouse pointer over the graph, and then hold down the middle mouse button. With the middle mouse button pressed down, click once on the left mouse button to zoom in and once on the right button to zoom out.

To pan the graph area:

- Drag left or right using the middle mouse button or use the scroll bar along the bottom and right side of the graph.

Using the Graph contextual menu for navigation:

There are several ways to navigate the graph area using the Spline Editor contextual menu as well.

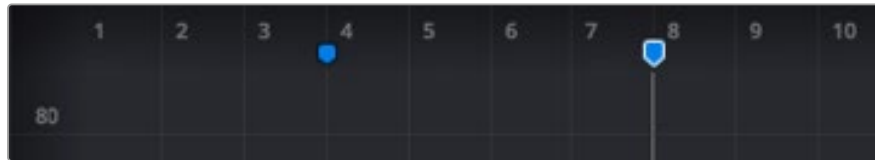
- Choose Scale > Scale to Fit (Command-F) to fit all active splines into the graph area.
- Choose Scale > Scale to Rectangle (Command-R) to draw a bounding box around the area of the graph you want centered and scaled. This has the same effect as clicking the Zoom to Rectangle button.
- Choose Scale > Default to reset the scaling of the graph area to default values.
- Choose Scale > Zoom In/Zoom Out to scale the graph area. This performs the same functions as pressing the + and - keys on the keyboard.
- Choose Scale > Auto Fit to scale the graph to fit all splines dynamically as you make splines visible and hidden. If the scaling is changed with Auto-Fit enabled, the graph area will scroll as you play the comp to view all the keyframes.
- Choose Scale > Auto Scroll to scroll the graph area if the splines fall outside the graph horizontally as you play.
- Choose Scale > Manual to disable all automatic attempts at showing splines in the graph.
- Choose Options > Fit Times to automatically scale along the X-axis to fit the selected spline. All visible splines are taken into account, not just the newly selected spline. With this option off, activating a new spline will not change the horizontal scale.
- Choose Options > Fit Values to automatically scale along the Y-axis to fit the selected spline. All visible splines are taken into account, not just the newly selected spline. With this option off, activating a new spline will not change the vertical scale.

Markers

Markers help identify important frames in a project. They may indicate a frame where a ray gun shoots a beam in the scene, the moment that someone passes through a portal in the image, or any other important event in the composite.

Markers added to the Timeline in the Cut, Edit, Fairlight, or Color page will appear in the Keyframes Editor and Spline Editor of the Fusion page. They can also be added from the Keyframes Editor or the Spline Editor while working in Fusion Studio or the Fusion page. Markers appear along the top of the horizontal axis Spline Editor's Time Ruler. They are displayed as small blue shapes, and when selected, a line extends from each guide down vertically through the graph.

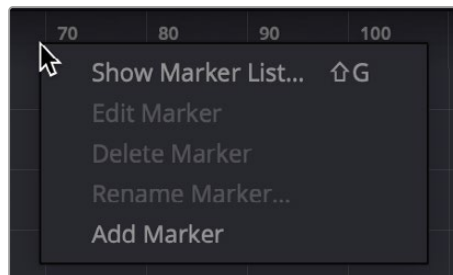
NOTE: Markers attached to clips in the Cut, Edit, Color, or Fairlight pages Timeline are not visible in Fusion's Spline Editor.



Unselected markers appear as blue shapes along the top, while selected markers display a vertical line running through the graph

Working with Markers

Markers call attention to a particular frame within a comp. They can be named, displayed in a list, and edited. After you add markers, you can easily jump the playhead between them, change their position, or delete them altogether.



Markers can be added by right-clicking in the horizontal time axis

To create a marker:

- Right-click in the horizontal axis Time Ruler and choose Add Marker.

To delete a marker, do one of the following:

- Drag the marker up outside the Spline Editor panel.
- Right-click on the marker and then choose Delete Marker from the menu.
- Select the marker and then press Delete or Backspace on the keyboard.
- From the Marker List, select a guide in the list and click the Del button.

To move a marker to a new frame, do one of the following:

- Drag the marker handle along the time axis.
- Right-click in the marker area and choose Options > Enable Marker Grab, and then drag the marker's vertical line to move the guide.

To move the Playhead to a marker:

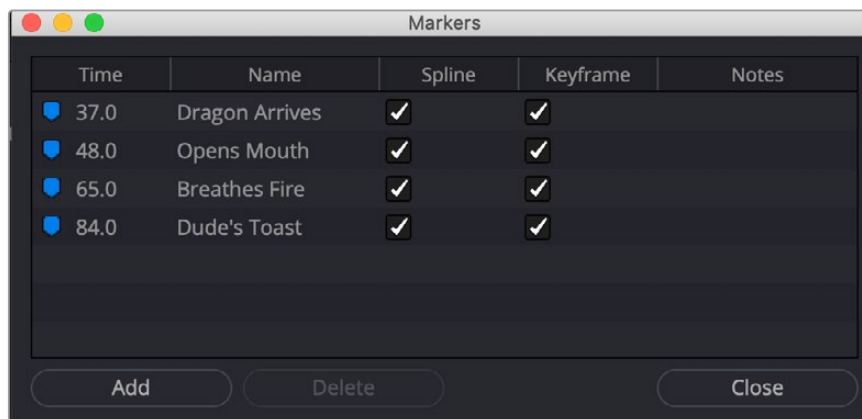
- Right-click on a marker and choose Set Current Time To [Frame number].

Using the Marker List

The Marker List is a list of all markers in the current comp. It can display the markers from either the Keyframes Editor, the Spline Editor, or both panels simultaneously. Clicking on the frame number or name of a guide causes the current time to change to that marker's frame. Since the Marker List is a floating window, it can remain open, allowing you to quickly jump to different markers while you work in the Spline Editor.

To show the Marker List:

- Right-click in the horizontal axis and choose Show Marker List, or press Shift-G.



The Marker List above shows markers in the current comp.

If markers currently exist in the comp, they are automatically displayed in the Marker List, regardless of whether they were added in the Keyframes Editor or the Spline Editor or any other page in DaVinci Resolve. You can also add markers directly from the Marker List, which can be helpful if you have multiple markers you need to add, and you know the rough timing.

To add a guide from the Marker List:

- 1 Click the Add button in the Marker List window.
- 2 Enter a frame number in the Time field.
- 3 Press Tab or click the Close button to close the Marker List.

To name a marker, do one of the following:

- In the Marker List, double-click in the Name column, to the right of the frame number, and enter a name for the marker.
- Right-click over a marker in the horizontal axis and choose Rename Marker. In the dialog that opens, enter a name for the marker.

Displaying Marker with the Timeline

The Marker List window includes checkboxes next to each marker that determines whether a marker displays in the Spline Editor, the Keyframes Editor, both, or neither. By default, when you create markers, they are active in both panels. To hide a marker from appearing in either panel, deselect the appropriate checkbox.

Autosnap

To assist in precisely positioning keyframe control points along the horizontal (time) axis, you can enable the Spline Editor's Autosnap function. Right-clicking over a spline and choosing Options > Autosnap Points provides a submenu with four options.

None: Allows free, sub-frame positioning of the keyframes.

Frame: Keyframes snap to the nearest frame.

Fields: Keyframes snap to the nearest field.

Markers: Keyframes snap to the nearest marker.

Autosnapping and Markers

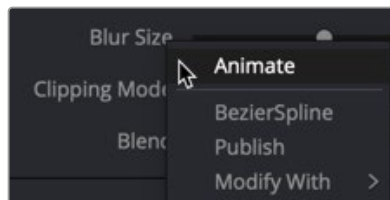
By default, a newly-created marker snaps to the closest frame. Moving markers with the mouse also snaps them to the current frame. You can change this behavior by selecting Options > Autosnap Markers > None or by selecting Options > Autosnap Markers > Field from the contextual menu.

Creating Animation Splines

Animation splines are created automatically when you keyframe a parameter in the Inspector or the Keyframes Editor. However, you can create an animation spline without first having to add a keyframe.

To create a spline:

- Right-click on the parameter to be animated in the Inspector, and choose Animate from the contextual menu.

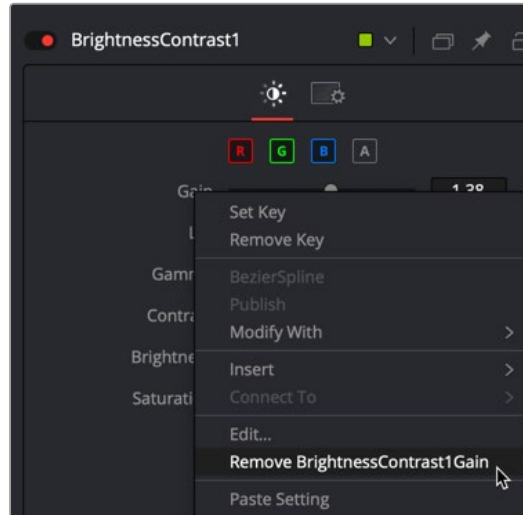


Choosing Animation from the menu displays a spline in the Spline Editor

Selecting Animate from the contextual menu connects the parameter to the default spline type. This is usually a Bézier Spline unless you change the default spline in the Defaults panel of the Fusion Preferences.

Deleting Animation Splines

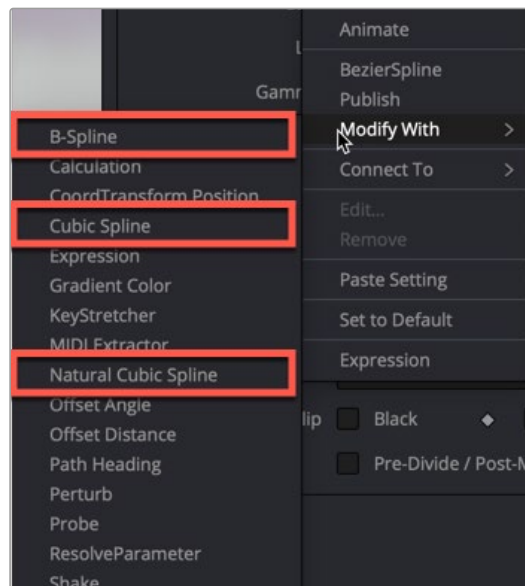
To remove an animation spline from a parameter, right-click on the control in the Inspector and select Remove [tool parameter's name] from the contextual menu. Removing a spline from a parameter only deletes the spline if no other tool in the composition is connected to the same spline at that time.



Removing a spline from a parameter removes the animation

Animating with Different Spline Types

A Bézier spline is the default spline unless changed in the Preferences. However, if you want to use a spline type other than Bézier for the animation curve, you can choose the spline type from the Modify With contextual submenu before creating any keyframes.



Three spline types in the Modify With menu

- **Bézier Spline:** Bézier splines are the default curve type. Three points for each keyframe on the spline determine the smoothness of the curve. The first point is the actual keyframe, representing the value at a given time. The other two points represent handles that determine how smoothly

the curve for the segments leading in and out of the keyframe are drawn. Bézier is the most used spline type because Bézier splines allow you to create combinations of curves and straight lines.



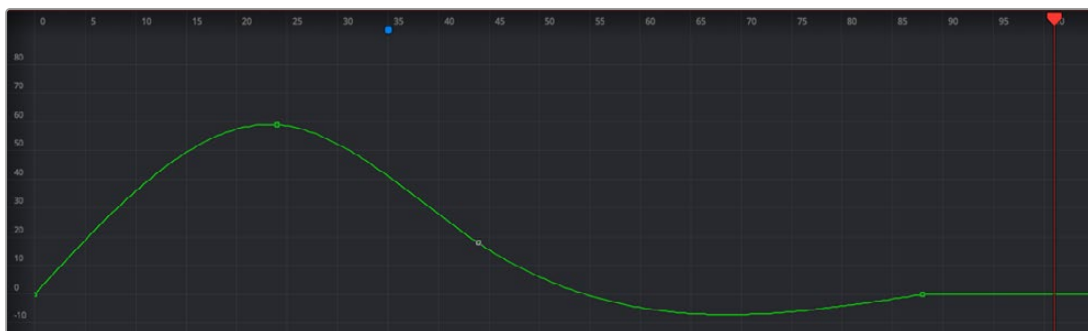
Bézier spline

- **Modify with > B-Spline:** B-splines use a single point to determine the smoothness of the curve. Instead of using handles, a single control point determines the value as well as the smoothness of the curve. Holding down the W key while dragging left or right on the control point adjusts the tension of the curve.



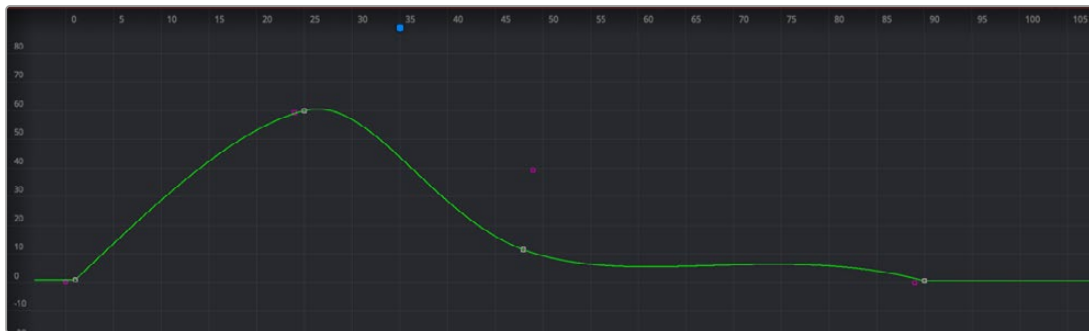
B-spline

- **Modify with > Cubic Spline:** Cubic splines are similar to Bézier splines, in that the spline passes through the control point. However, Cubic splines do not display handles and always make the smoothest possible curve. In this way, they are similar to B-splines. This spline type is almost never used.



Cubic spline

- **Modify with > Natural Cubic Spline:** Natural Cubic splines are similar to Cubic splines, except that they change in a more localized area. Changing one control point does not affect other tangents beyond the next or previous control points.



Natural Cubic spline

Working with Keyframes and Splines

Once you animate a parameter and display the Spline Editor, you can manipulate the spline's keyframes (and thus the animation) in a variety of ways. By selecting Keyframe control points, you can move, copy, and change the interpolation of your animation.

Adding Keyframes

Once you create one keyframe, additional keyframes are automatically added to a spline whenever you move the playhead and change the value of that spline's parameter. For example, if you change the strength of an animated glow at frame 15, a keyframe with the new value occurs on frame 15.

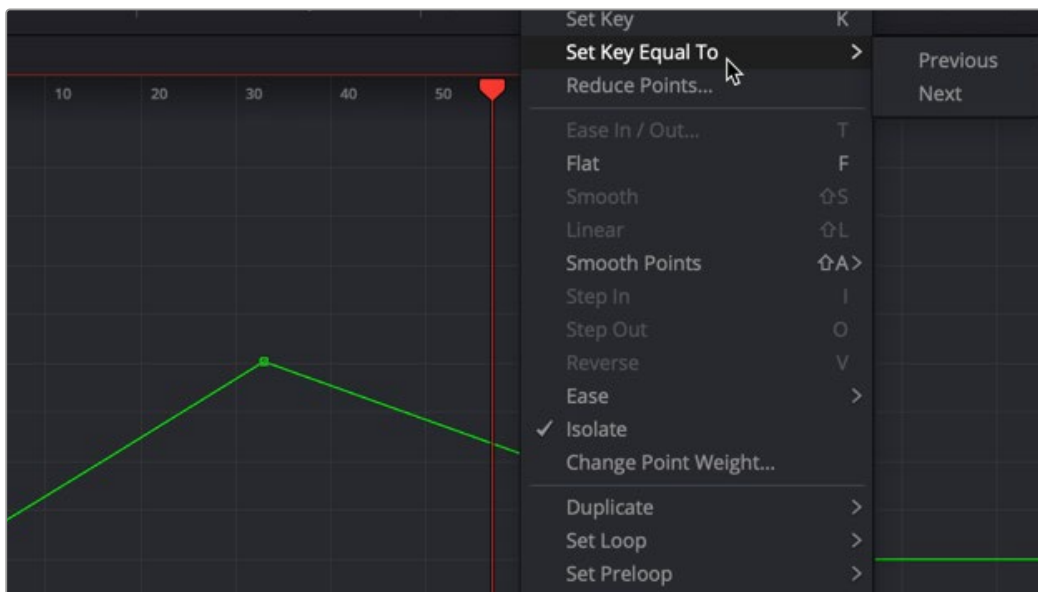
In the Spline Editor, control points can also be added directly to a spline by clicking on the spline where you want to add the new keyframe.

Adding Keyframes at the Playhead

If you want to add a new keyframe at the current playhead location, pressing Command-K on the keyboard or right-clicking in the graph and choosing Set Key adds a keyframe under the playhead.

Adding Equal Keyframes

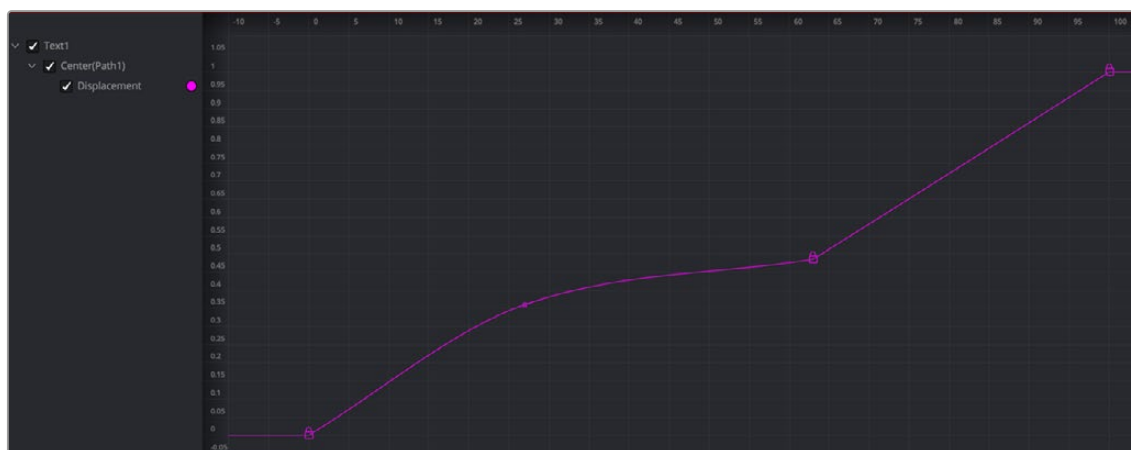
If you want to hold a value over several frames, right-clicking in the graph area and choosing Set Key Equal To displays a submenu to add a new keyframe with a value equal to the next or the previous keyframe.



To hold a value between two keyframes choose from the Set Key Equal To submenu

Locked and Unlocked Controls Points

When animating the Center X/Y or Pivot X/Y parameters on any tool, you create a displacement spline in the Spline Editor. The displacement spline represents the relative offset position of the animated object along its path. Since the displacement spline is relative, keyframes use a value between 0.0 and 1.0. A displacement value of 0.0 in the Spline Editor indicates that the object is at the very beginning of a path. A value of 1.0 indicates that the object is positioned at the end of the path.



The displacement spline represents the relative position along a motion path

Displacement paths are composed of locked and unlocked points. Whether a point is locked is determined by how you added it to the polyline. Locked points on the spline have an associated point in the viewer's motion path; unlocked points do not have a corresponding point in the viewer's motion path. Each has a distinct behavior, as described below.

TIP: You can convert displacement splines to X and Y coordinates by right-clicking over the motion path in the viewer and choosing Path#: Polyline > Convert to X/Y Path.

Locked Points

Locked points are the motion path equivalents of keyframes. They are created by moving the playhead position and changing the parameter value. These points indicate that the animated object must be in a specified position on a specified frame. Since these keyframes are only related to position along the path, they can only be moved horizontally along the spline's time axis.

The locked points appear as larger-sized lock icons in the Spline Editor. Each locked key has an associated point on the motion path in the viewer.

Deleting a locked point changes the overall timing of the motion.

Unlocked Points

Unlocked points are created by clicking directly on the spline in the Spline Editor. These points give additional control over the acceleration along the motion path without adjusting the path itself. Conversely, you can add unlocked points in the viewer to control the shape of the motion path without changing the timing.

You can change an unlocked point into a locked point, and vice versa, by selecting the point(s), right-clicking, and choosing Lock Point from the contextual menu.

For more information on motion paths and locked keyframes, see Chapters 71 and 73 in the DaVinci Resolve manual or Chapters 8 and 10 in the Fusion Studio manual.

Selecting, Moving, and Deleting Keyframes

The placement of keyframes greatly affects the style of the animation. Using the graph, you can select keyframes and move them up or down to change their value or move them left and right to change the timing. Keyframes can be copied and pasted between splines and parameters.

Methods of selecting keyframes:

- Click directly on a keyframe on the spline, or drag a bounding box around the keyframe.
- Drag a bounding box that encompasses multiple keyframes to select more than one.
- To add or remove a keyframe from the current selection, hold down the Command key while selecting the keyframes. This will remove currently selected keyframes and add currently unselected keyframes.
- Press Command-A or right-click in the graph area and choose Select Points > Select All from the contextual menu to select all keyframes from the active splines.

Moving Keyframes

You can freely move keyframes with the mouse, keyboard, or the edit point controls. Keyframes can even pass over existing points as you move them. For instance, if a keyframe exists on frame 5 and frame 10, the keyframe at frame 5 can be repositioned to frame 15.

To move keyframes with the mouse:

- Drag the selected keyframe to its new position in the graph. If more than one keyframe is selected, all selected keyframes will be moved simultaneously.
- Hold down the Option key before dragging the keyframe to constrain its motion to a single axis.

To move keyframes with the keyboard:

- The Up and Down Arrow keys will adjust the value of the keyframes by a small amount.
- The current scale of the graph determines the degree of vertical movement applied to the value with each key press. The closer the zoom in the spline, the finer the adjustment.
- Hold down the Shift key while pressing the Up or Down Arrow keys to increase the value adjustment in larger increments.

To move keyframes using edit fields:

- The Value and Time Editors are found on the far right of the toolbar. These number fields allow explicit values to be entered for selected keyframes. These controls are explained in more depth later in this chapter.

To delete one or more keyframes:

- Select one or more keyframes and press the Delete or Backspace key on the keyboard. This only removes the keyframes; it does not remove the spline, even if there are no keyframes on the spline. To remove the spline, right-click over the parameter in the Inspector and choose Remove [parameter name].

Showing Key Markers

You can adjust the position of the keyframes in time, without worrying about manipulating splines, by using the key markers. The horizontal time axis can show markers that indicate the position of each keyframe. The display of these markers is enabled by right-clicking in the graph and choosing Show > Key Markers from the contextual menu, or by clicking on the Show Key Markers button in the toolbar.



The key markers show keyframes in the horizontal axis using the same color as the splines

Copying and Pasting Keyframes

To precisely match animation, Keyframes can be copied to a new location on the same spline or onto completely different splines and different tools.

There are two options in the graph's contextual menu for copying keyframes. Choosing Copy Points (Command-C) copies all selected points. Choosing Copy Value copies a single point identified by the pointer from multiple selected points. This does not deselect your selection set, and you can pick out numbers as needed.

To copy and paste points to a new location on the same spline:

- 1 Select the desired keyframes on the spline.
- 2 Right-click over the spline and choose Copy Points from the contextual menu or press Command-C.
- 3 Click in an empty area of the graph to deselect all the copied points.
- 4 Move the playhead to the area of the spline where you want the points pasted and press Command-V.

Or, move the pointer over the spline where you want the points pasted, and when the spline highlights, right-click and choose Paste Points/Value.

Alternatively, you can copy and paste keyframes by dragging them with the mouse. After you select the points, hold down the Command key and drag the points along the spline to where you want them pasted.

To copy and paste keyframes from one spline to another:

- 1 Make one spline the active visible spline and select the desired keyframes on the spline.
- 2 Right-click over the spline and choose Copy Points from the contextual menu or press Command-C.
- 3 Set the spline to viewed or disabled using the status checkbox next to the spline's name in the header.
- 4 Make the destination spline the active visible spline and select the keyframe on the spline where the new keyframes should be pasted.
- 5 Right-click and choose Paste Points/Value or press Command-V.

You can copy a single point's value from a group of selected points. Since this process does not deselect the selected set, you can continue picking out values as needed without having to reselect points.

To copy and paste a keyframe value:

- 1 Make one spline the active visible spline and select all the keyframes on the spline.
- 2 Right-click over a single point and choose Copy Value from the contextual menu.
- 3 Set the spline to viewed or disabled using the status checkbox next to the spline's name in the header.
- 4 Make the destination spline the active visible spline and select the keyframe on the spline where the new keyframe should be pasted.
- 5 Right-click and choose Paste Points/Value or press Command-V.

Keyframes can also be pasted with an offset, allowing you to duplicate a spline shape but increase the values or shift the timing using an offset to X or Y.

To paste keyframes points and values with an X or Y offset:

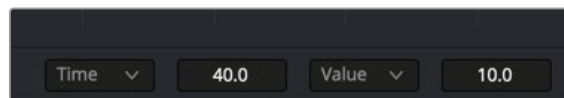
- 1 Make one spline the active visible spline and select the desired keyframes on the spline.
- 2 Right-click over the spline and choose Copy Points from the contextual menu or press Command-C.
- 3 Set the spline to viewed or disabled using the status checkbox next to the spline's name in the header.

- 4 Make the destination spline the active visible spline and select the keyframe on the spline where the new keyframes should be pasted.
- 5 Right-click and choose Paste with Offset. In the Offset dialog, enter the Y value, which will be added to the values of the pasted keyframes.

TIP: You cannot copy and paste between different spline types. For instance, you cannot copy from a Bézier spline and paste into a B-spline.

Time and Value Editors

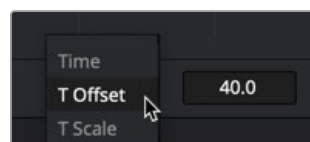
The Time and Value Editors in the lower-right corner of the Spline Editor are used to change the position and parameter value of a keyframe by entering a number into the number field for each button. Each field can switch between three modes that help modify the time and value of a keyframe in three precise but distinct ways. The default mode for each field takes the explicit frame number or parameter value at which you want the keyframe set. The other modes offset and scale the keyframe's position or value.



Use the number fields to enter in a value or a specific time to change the selected keyframe

Time Editor

The Time Editor is used to modify the current time of the selected keyframe. You can change the Time mode to enter a specific frame number, an offset from the current frame, or spread the keyframes based on the distance (scale) from the playhead. You can select one of the three modes from the Time mode drop-down menu.



Three time editing modes are selectable from the Time mode drop-down menu

Time

The number field shows the current frame number of the selected control point. Entering a new frame number into the field moves the selected control point to the specified frame. If no keyframes are selected or if multiple keyframes are selected, the field is empty, and you cannot enter a time.

Time Offset

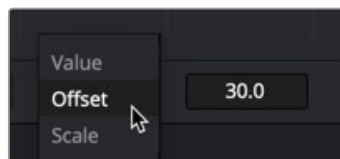
Selecting T Offset from the drop-down menu changes the mode of the number field to Time Offset. In this mode, the number field offsets the selected keyframes positively or negatively in time. An offset of either positive or negative values can be entered. For example, entering an offset of 2 moves a selected keyframe from frame 10 to 12. If multiple keyframes were selected in the previous example, all the keyframes would move two frames forward from their current positions.

Time Scale

Selecting T Scale from the drop-down menu changes the mode of the number field to Time Scale. In this mode, the selected keyframes' positions are scaled based on the position of the playhead. For example, if a keyframe is on frame 10 and the playhead is on frame 5, entering a scale of 2 moves the keyframe 10 frames forward from the playhead's position, to frame 15. Keyframes on the left side of the playhead would be scaled using negative values.

Value Editor

The Value Editor is used to modify the selected keyframe's parameter value using one of three Value modes. You can change the Value mode to enter a specific value for a parameter, an offset from the value, or to spread the values. The mode is chosen from the Value mode drop-down menu.



Three value editing modes are selectable from the Value mode drop-down menu

Value

The number field shows the value of the currently selected keyframes. Entering a new number into the field changes the value of the selected keyframe. If more than one keyframe is selected, the displayed value is an average of the keyframes, but entering a new value will cause all keyframes to adopt that value.

Value Offset

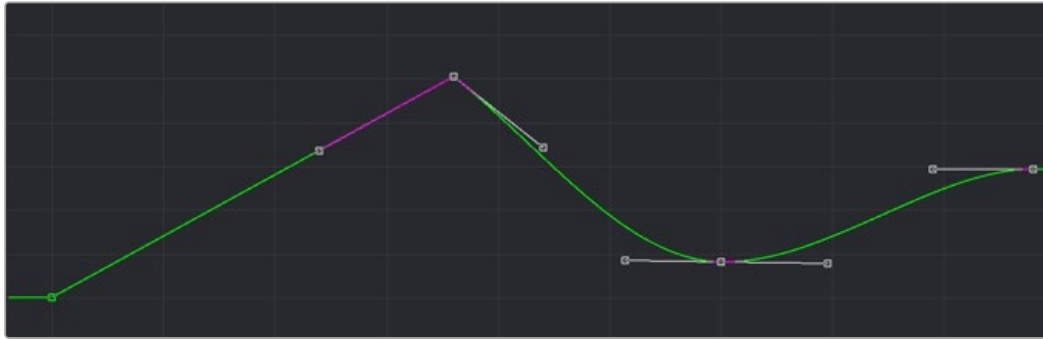
Choosing Offset from the drop-down menu sets the Value Editor to the Offset mode. In this mode, the value for the selected keyframes are offset positively or negatively. An offset of either positive or negative values can be entered. For example, entering a value of -2 changes a value from 10 to 8. If multiple keyframes are selected, all the keyframes have their values modified by -2.

Value Scale

Choosing Offset from the drop-down menu sets the Value Editor to the Scale mode. Entering a new value causes the selected keyframes' values to be scaled or multiplied by the specified amount. For example, entering a value of 0.5 changes a keyframe's value from 10 to 5.

Modifying Spline Handles

All Bézier spline keyframes have a pair of control handles to shape the spline as it passes through the key point. These handles are only displayed when the keyframe is selected. Initially, these handles are set to linear, creating straight line changes between keyframes. However, any control point can be made smooth by right-clicking over it and choosing Smooth or pressing Shift-S.



Bézier splines can mix linear and smooth curves

Dragging on a keyframe's handles adjusts the slope of the segments passing through the spline. By default, the two control handles on a control point are locked together so that if one moves, the one on the other side moves with it. This maintains a constant tension through the keyframe. There are situations, however, when it is desirable to modify these control handles separately for a more pronounced curve or effect.

To temporarily break the control handles on a Bézier spline, moving one independently of the other:

- 1 Select the control point to be modified.
- 2 Hold down the Command key and drag one of the control handles. They will now move independently of each other, as long as the Command key is held down.

To treat all Bézier handles as independent in the Spline Editor:

Right-click in the graph and choose Independent Handles from the Options contextual menu.

Enabling this option causes all the Bézier handles to be independent. This is the same as using the Command key when moving a handle, except it is applied to all control points until it is disabled.

Reducing Points

When there are too many control points too close together on a spline, you can choose Reduce Points to decrease their number, making it easier to modify the remaining points. The overall shape of the spline is maintained as closely as possible while eliminating redundant points from the path.

To reduce the number of points on a spline:

- 1 Select the range of keyframes you want to thin out.
- 2 Right-click in the graph area and choose Reduce Points from the contextual menu.
- 3 When the Reduce Points dialog appears, drag the slider to a lower value.

You can set the slider value as low as possible as long as the spline still closely resembles the shape of your original spline.

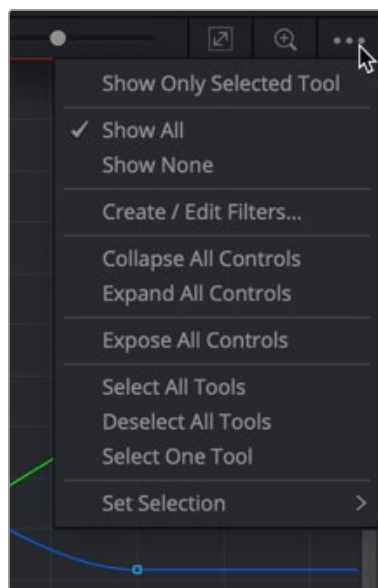
TIP: When the value is 100, no points will be removed from the spline. Use smaller values to eliminate more points.

Filtering the Spline Editor

The animation splines for multiple parameters can be displayed within the Spline Editor simultaneously, and Fusion offers several ways for you to choose which spline to view and which to edit.

A complex composition can easily contain dozens, if not hundreds, of animation curves. As a composition grows, locating a specific spline can become more difficult. There are two ways to filter the splines shown in the Spline Editor: display selected tools only or create a filter to show only certain tools.

The Spline Editor includes different ways to control which splines are displayed. The majority of these options are available in the Options menu, located in the upper-right corner of the Spline Editor panel.



The Options menu is used to control which splines are displayed in the Spline Editor

Show Only Selected Tool: You can choose to limit the splines displayed in the Spline Editor by showing only the splines from selected tools. Choosing this option at the top of the Options menu displays only the splines for tools currently selected in the Node Editor.

Show All/None: The default behavior of the Spline Editor displays all the splines for all the nodes with animated parameters. You can override this by enabling Show Only Selected Tools in the Options menu. You can also disable the Show All setting by choosing Show None, in which case the Spline Editor remains empty.

Expose All Controls: The Expose All Controls option is a way of not filtering the parameters. Choosing this option displays all parameters in the Spline Editor header for all nodes in the Node Editor. It can be a fast way of activating one of the parameters and automatically adding an animation spline for it if one does not exist.

With a large number of nodes displayed, which themselves might have a large number of parameters, this might lead to cluttering and slowing down the interface. This option is most effective when used in conjunction with the Show Only Selected Tool option to limit the number of nodes and parameters displayed and yield optimum performance.

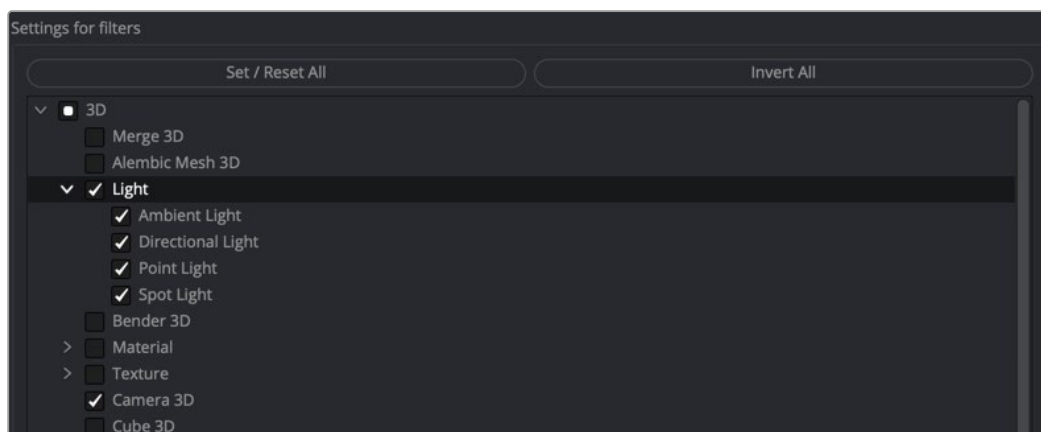
Follow Active: The Follow Active option is located by right-clicking in the graph and choosing Options > Follow Active. This option provides a way to filter the splines in the graph while not filtering the header list of tools. Where the Show Only Selected Tool option hides other tools in the header, the Follow Active option leaves the header displaying all the tools but automatically enables only the splines of the Active tool.

Working with Filters

Filters allow you to select the specific types and classes of tools shown in the Spline Editor and Keyframes Editor. For example, you can make a filter that shows only particle nodes or one that only shows color correction and brightness/contrast tools.

To create a filter:

- 1 From the Options menu, choose Create/Edit Filters.
- 2 Click the New button to create a new filter and name the new filter in the dialog box.
- 3 Enable a checkbox next to the entire category or the individual tools in each category to determine the tools included in the filter.



Enable each tool you want to keep in the Spline Editor when the filter is selected

The Invert All and Set/Reset All buttons can apply global changes to all the checkboxes, toggling the selected states as described.

To switch the selection state of the categories when creating a filter list:

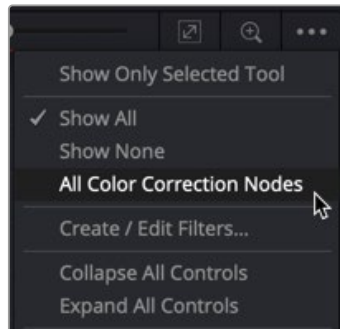
- 1 Click the Invert All button.
- 2 After configuring the custom filter, click the Save button to close the Settings dialog and save the filter.

To enable all checkboxes or disable all checkboxes:

- 1 Click the Set/Reset All button as many times as needed until all categories are either checked or unchecked.
- 2 After configuring the custom filter, click the Save button to close the Settings dialog and save the filter.

To apply a filter to the Spline Editor:

- Choose the desired filter by name from the Options menu. The filter applies to both the Spline Editor and the Timeline.



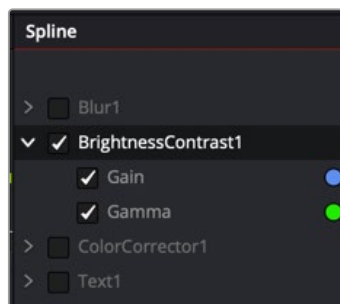
Each filter you create is listed in the Options menu

To disable a filter and show all tools in the Spline Editor again:

- Choose Show All from the Options menu.

Changing a Spline's Status

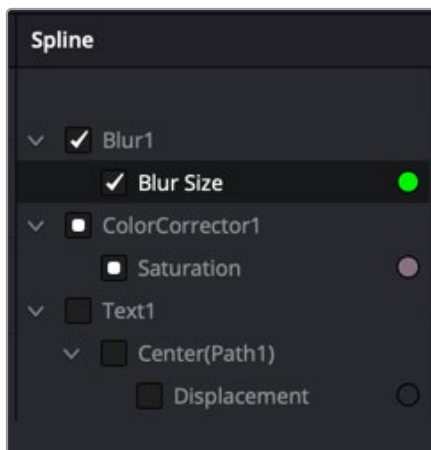
The Spline header is a hierarchical list of animated parameters and their parent nodes. Clicking the disclosure arrow next to a tool's name reveals all the names of the animated parameters on that tool. Clicking directly on the parameter name in the Spline header activates that spline for display and editing.



The Spline header with the Brightness Contrast tool and its animated Gain and Gamma parameters

Tool Status Checkbox

Next to the name of each spline is a checkbox that indicates the spline's status. When you select a parameter name, the checkbox becomes active, allowing you to see and edit the spline in the graph. There are three selection modes for each checkbox: active, viewed, and disabled. Clicking directly on the checkbox will toggle it between these three states. Changing the state of the parent node checkbox sets the state for all splines for that node.



Active: When the checkbox is enabled with a check mark, the spline is displayed in the graph and can be edited.

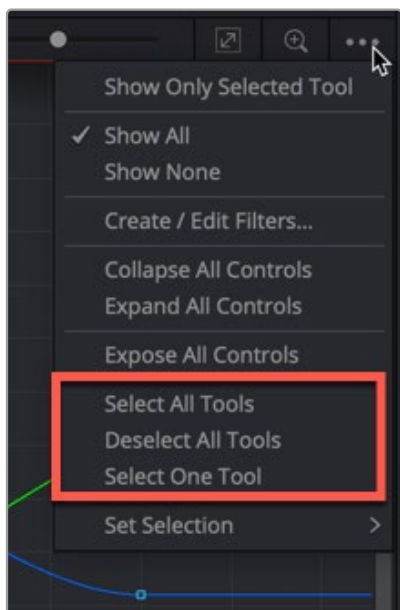
Viewed: When the checkbox is enabled with a solid gray box, the spline is visible in the graph but cannot be edited. It is read-only.

Disabled: When the checkbox is clear, the spline is not visible in the graph and cannot be edited.

Checkboxes determine which splines are visible and editable

Selection States

There are three selection options, labeled Select All Tools, Deselect All Tools, and Select One Tool, that determine how the items in the Spline Editor header behave when a checkbox or label is selected to activate a spline. These states are located in the Options menu in the upper-right corner of the Spline Editor.



Select All Tools: Choosing this option activates all splines for editing.

Deselect All Tools: Choosing this option sets all spline checkboxes to disabled.

Select One Tool: This option is a toggle. When Select One Tool is chosen from the menu, only one spline in the header is active and visible at a time. Clicking on any spline's checkbox will set it to active, and all other splines will be cleared. When disabled, multiple splines can be active in the header.

The Options menu selection states make it easier to select and deselect all parameters in the header

Selection Groups

It is possible to save the current selection state of the splines in the header, making selection groups that can easily be reapplied when needed. To create a selection group, right-click over any parameter in the header or in an empty area of the graph and choose Save Current Selection from the contextual menu. A dialog will appear to name the new selection.

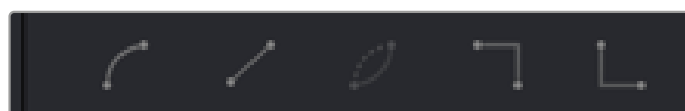
To reapply the selection group, choose the selection group by name from the Set Selection menu in the same contextual menu. Other context menu options allow selection groups to be renamed or deleted.

Reshaping Splines Using the Toolbar

There are several ways to manipulate the shape of a spline, thereby altering the animation that spline generates. Other than manually adjusting Bézier handles, you can quickly squish, stretch, loop, and reverse a spline. You can also quickly change the interpolation between keyframes from the default linear motion to more natural smooth motion. All these options are provided at the bottom of the Spline Editor in the toolbar. The toolbar is divided into different groups for setting interpolation, reversing splines, looping splines, time stretching, and reshaping splines.

Interpolation

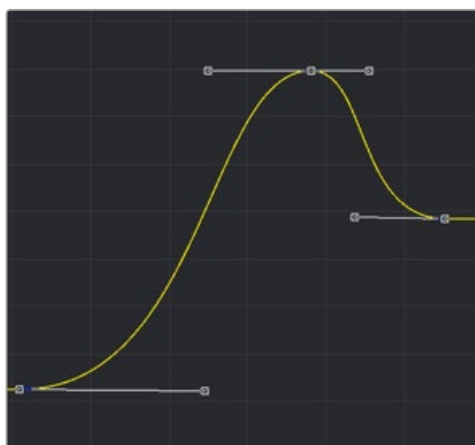
Keyframes are specific frames in an animation where control points are set to exact values on a given parameter. Interpolation is the method used to fill in the unknown values between two keyframes. Fusion automatically interpolates between two keyframes. However, you may want to modify the interpolation to achieve a specific style of animation. The Spline Editor includes several interpolation methods you can choose from using the toolbar.



Interpolation buttons in the toolbar: Smooth, Linear, Invert, Step In, and Step Out

Smooth

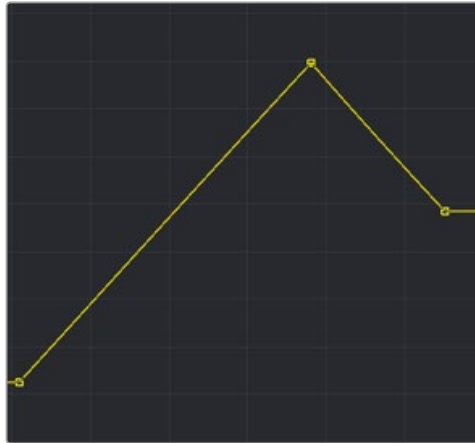
A smoothed segment provides a gentle keyframe transition in and out of the keyframe by slightly extending the direction handles on the curve. This slows down the animation as you pass through the keyframe. To smooth the selected keyframe(s), press Shift-S or click the toolbar's Smooth button.



Smooth interpolation between keyframes

Linear

A linear segment effectively takes the shortest route between two control points, which is a straight line. To make the selected keyframe(s) linear, press Shift-L or click the Linear button in the toolbar.



Linear interpolation between keyframes

TIP: Invert is used only for non-animated LUT splines, which are currently only available in the LUT Editor window.

Step In/Step Out

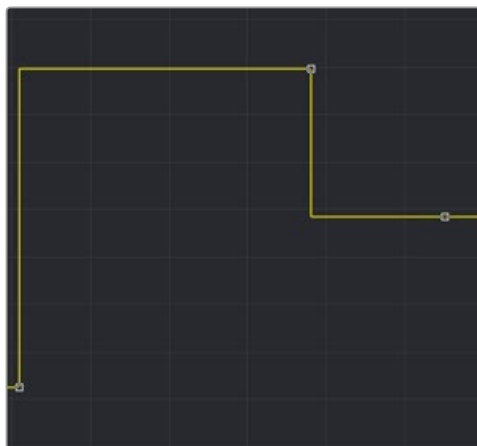
On occasion, it is not desirable to have any interpolation between two keyframes. Instead, the value of one keyframe may hold its value until another keyframe changes it. For these cases, use the Step In or Step Out mode.

Step In causes the value of the previous keyframe to hold, then jump straight to the value of the next keyframe.



Step In holds a value until the next keyframe is reached in the comp

Step Out causes the value of the selected keyframe to hold right up to the next keyframe.



Step Out switches immediately to the next keyframe value in a comp

Step In and Step Out modes can be set for selected keyframes by clicking on the toolbar buttons for each mode, or by right-clicking and choosing the appropriate option from the contextual menu. The keyboard shortcuts I and O can also be used to enable Step In and Step Out on selected keyframes.

Reversing Splines

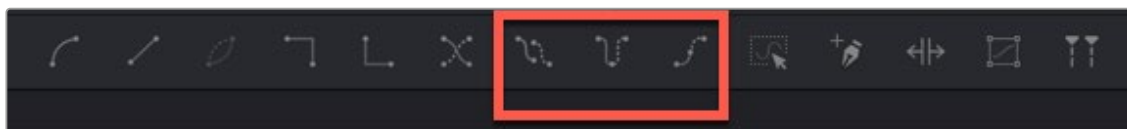
Reverse inverts the horizontal direction of a segment of an animation spline. To apply reverse, choose a group of points in a spline and click the Reverse button, or right-click and choose Reverse from the contextual menu, or press the V key. The group of points is immediately mirrored horizontally in the graph. Points surrounding the reversed selection may also be affected.



The Reverse button in the toolbar

Looping Splines

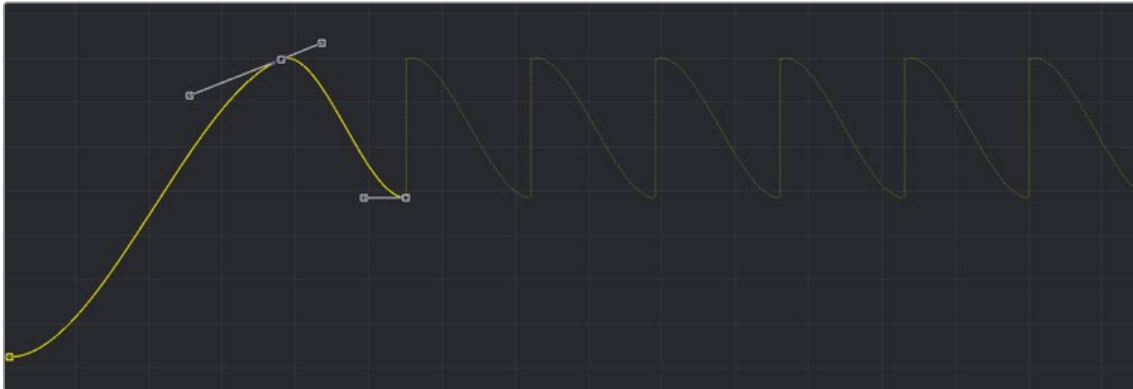
It is often useful to repeat an animated section, either infinitely or for a specified number of times, such as is required to create a strobing light or a spinning wheel. Fusion offers a variety of ways to repeat a selected segment.



The various Loop buttons in the toolbar

Set Loop

To repeat or loop a selected spline segment, select the keyframes to be looped. Select Set Loop from the contextual menu or click on the Set Loop button in the toolbar. The selected section of the spline repeats forward in time until the end of the global range, or until another keyframe ends the repeating segment.



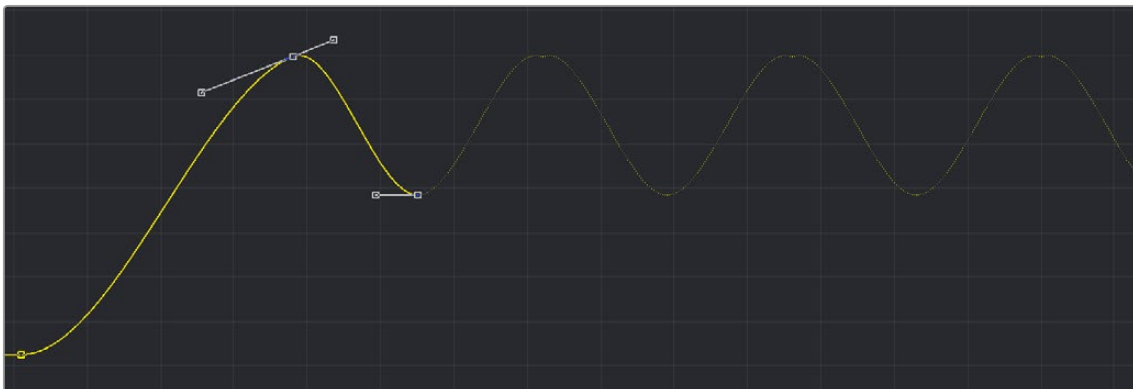
A looped section in the graph

Changing and Removing the Loop

You can change the looped segment by modifying any of the keyframes or control points originally used to create the loop. Simply select one of the originating key points, make any necessary modifications, and the looped segment updates. To remove the loop, select the keyframes you used to create the loop, and then click the Loop button in the toolbar.

Ping-Pong

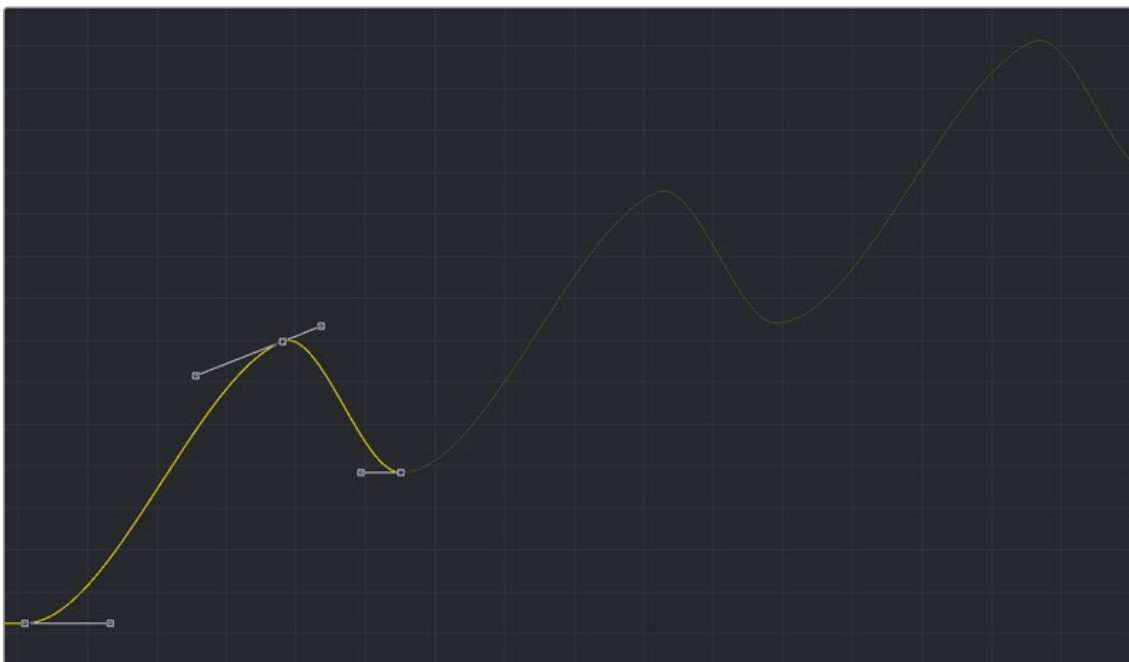
The Ping-Pong Loop mode repeats the selected segment, reverses each successive loop, and then repeats. Ping-pong looping can be enabled on the selected segments from the context menu or the toolbar.



A ping-pong section in the graph

Relative Loop

The Relative Loop mode repeats the segment like the Loop, but each repetition adds upon the last point of the previous loop so that the values increase steadily over time.



A Relative Loop section in the graph

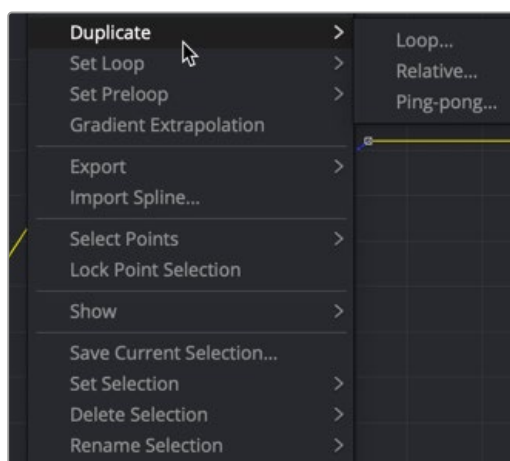
Looping Backward

You can choose Set Pre-Loop by right-clicking in the graph area and choosing it from the contextual menu. This option contains the same options for looping as the Loop option buttons in the toolbar, except that the selected segment is repeated backward in time rather than forward.

Repeating Splines X Number of Times

You can duplicate splines and repeat them a set number of times by right-clicking in the graph area and choosing Duplicate from the contextual menu. Duplicated splines are like looped splines, except that the selected segment repeats only a specified number of times, and each repetition is a copy rather than an instance. Adjustments to the original segment do not alter the shape of its repetitions.

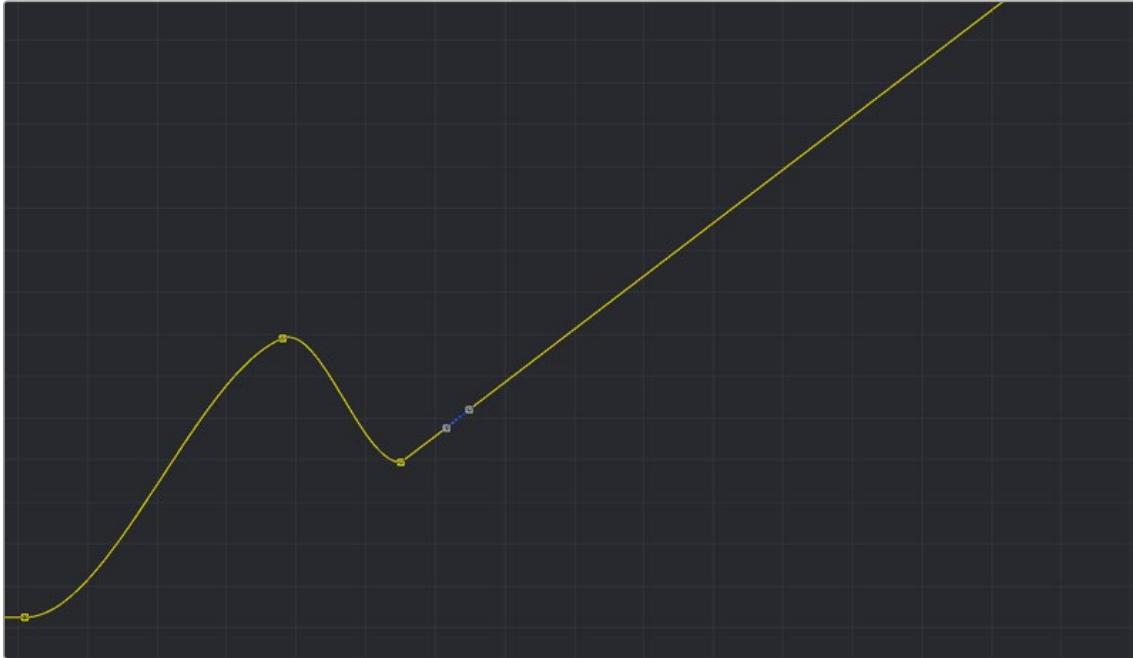
The Duplicate modes are only accessed from the Duplicate contextual menu, which reveals a submenu with all the looping modes described above. Selecting any of these modes opens a dialog in which the number of repetitions can be entered.



The Duplicate contextual submenu

Gradient Extrapolation

You can choose Gradient Extrapolation by right-clicking in the graph area and choosing it from the contextual menu. This option continues the trajectory of the last two keyframes.



The Gradient Extrapolation applied to the spline

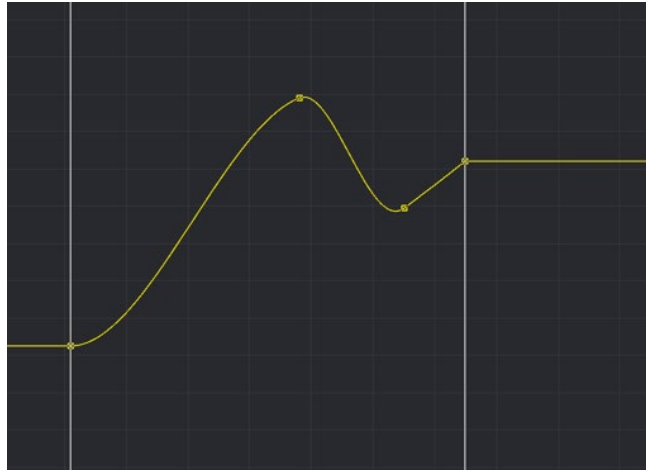
Time Stretching

Time Stretching allows for a selected group of keyframes to be proportionally stretched or squashed. This allows you to change the duration of the animation while keeping the relative distance between each keyframe. To enable spline stretching, select the group of keyframes that you want to time stretch, and then choose Modes > Time Stretching from the graph's contextual menu or click the Time Stretch button in the toolbar.



The Time Stretch button in the toolbar

When you have more than one keyframe selected on the spline, enabling Time Stretch surrounds the outer keyframes with two vertical white bars. Drag on the white vertical bars to stretch or shrink the timing of the spline segments within the bars. Drag these bars back and forth to stretch or squash the spline segment.



The Time Stretch bars in the graph

TIP: If no keyframes are selected when you enable Time Stretch, drag a rectangle to set the boundaries of the Time Stretch.

To disable the Time Stretching mode:

- Click on the Time Stretch button in the toolbar again or reselect Modes > Time Stretching from the contextual menu.

Shape Box

The Shape Box transform mode is similar to Time Stretching; however, it can adjust the vertical scaling of keyframe values as well as time.

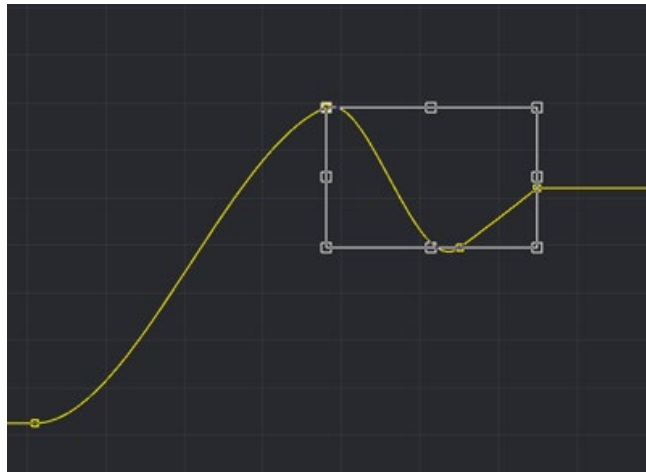
To enable the Shape Box, do one of the following:

- Select a group of keyframes, and then choose Modes > Shape Box from the contextual menu.
- Select a group of keyframes, and then click the Shape Box button in the toolbar.
- Select a group of keyframes, and then press Shift-B to enable or disable the Shape Box mode.



The Shape Box button in the toolbar

A white rectangle outlines the selected points when the mode is enabled. To scale, skew, or stretch the spline, drag on any of the control points located around the box. To move all the keyframes, drag on the box edges.



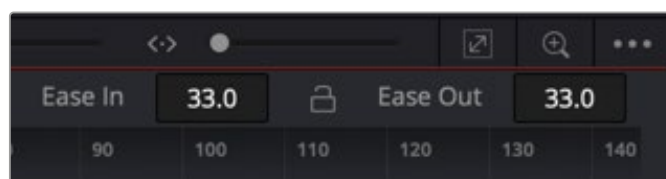
The Shape Box in the graph

TIP: If no points are selected, or if you want to select a new group of keyframes, you can drag out a new rectangle at any time.

Ease In/Out

For a more precise way to adjust the length of Bézier direction handles attached to selected keyframes, you can use the Spline Ease dialog. To show the dialog, select a keyframe in the graph, and then choose Edit > Ease In/Out from the graph's contextual menu or press T on the keyboard.

The Ease In/Out controls appear above the graph area. You can drag over the number fields to adjust the length of the direction handles or enter a value in the fields.



The Ease In/Out controls above the graph area

Clicking the Lock In/Out button will collapse the two sliders into one, so any adjustments apply to both direction handles.

Importing and Exporting Splines

Spline shapes can be imported and exported from or to an ASCII text file. This makes it easier to save complex spline curves for later reuse, or to transfer tracking, path, and animation data from one application to another. Exported splines are assigned the file extension .spl for easy identification.

To export a spline:

- 1 Select the active spline in the Spline Editor.
- 2 Right-click on the spline in the graph area to display the contextual menu and select Export.
- 3 Choose from three format options in the submenu.
- 4 Enter a name and location in the file browser dialog, and then click Save.

Exporting a spline gives you three options. You can export the Samples, Key Points, or All Points. Samples adds a control point at every frame to create an accurate representation of the spline. Key Points replicates the control point positions and values on the spline using linear interpolation. All Points exports the spline as you see it in the Spline Editor, using the same position, value, and interpolation.

To import a spline:

- 1 Add an animation spline for the parameter.
- 2 In the Spline Editor, right-click on the animation spline and select Import Spline from the contextual menu.
- 3 In the File Browser dialog, select the spline curve .spl file, and then click Open.

Importing a new curve will replace any existing animation on the selected spline.